

Ahan Bhatt

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EDUCATION

University of California, Santa Cruz [3.94 GPA]

Master of Science in Computer Science and Engineering

Santa Cruz, California

Aug. 2025 – Jun 2026

Indus Institute of Technology and Engineering [3.94 GPA]

Bachelor of Technology in Computer Engineering

Ahmedabad, India

Aug. 2020 – July 2024

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R

Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas, Matplotlib

LLM and Data Tools: LangChain, ChromaDB, RAG, vector databases, PyPDF

Backend and Deployment: Node.js, FastAPI, Flask, REST APIs, Docker, Kubernetes, Linux, Git, CI/CD, Nginx

Cloud: AWS EC2, AWS S3, AWS IAM, AWS Lambda, SageMaker, CloudWatch

Data Engineering: PostgreSQL, Redis, ETL, SQL optimization, schema design, data cleaning, time-series preprocessing

Software Engineering: TDD, API integration, version control, DSA, System Design, Performance Optimization, Code Reviews, Debugging

EXPERIENCE

Researcher

AI Explainability and Accountability (AIEA) Lab @ UCSC

September 2025 – June 2026

Santa Cruz, California

- Built ReliaGuard Studio with Next.js, TypeScript, FastAPI, PostgreSQL, Docker, and MLflow to detect over/underreliance in human-AI decisions using neuro-symbolic risk modeling, conformal gating, and explainable dashboards.

Data Scientist

AI Institute of South Carolina

June 2024 – May 2025

Columbia, South Carolina (remote)

- Conducted research on multimodal toxicity mitigation and anomaly detection, using PostgreSQL-backed experiment tracking to manage dataset metadata, model outputs, evaluation metrics, and benchmarking results for projects like DE-HATE and Time Series Foundation Model evaluation.

Data Scientist

Space Applications Centre, ISRO

Jan. 2024 – May. 2024

Ahmedabad, Gujarat

- Built an end-to-end LSTM satellite clock-bias prediction platform with Apache Kafka live streaming and a Next.js/React/TypeScript UI, automating preprocessing and anomaly filtering to improve navigation timing forecasts.

Software Engineer (AI)

Indus Institute of Technology

May 2022 – Jan. 2024

Ahmedabad, Gujarat

- Developed Node.js-backed LLM pipelines for a university website chatbot, automated knowledge graph generation, and retrieval-augmented medical QA, evaluating GPT-4, LLaMA-2, and BERT for structured semantic extraction and grounded reasoning.

PROJECTS

[Research Forge – A Self-Improving Research Agent](#)

- Working on Research Forge, a universal self-improving AI research agent that autonomously discovers papers, generates hypotheses, runs experiments, and updates its reasoning strategy using long-term graph memory.

[Meeting AI Assistant](#)

- Built a real-time AI meeting copilot desktop app (Electron + React + Express) that captures system audio, transcribes conversations, and streams context-aware responses using OpenAI API with screenshot and document context support.

NeuroSymbolic Cognitive Decline Analyzer

- Developed a hybrid NeuroSymbolic AI framework combining neural feature extraction with symbolic reasoning and fuzzy logic rules to assess cognitive decline risk associated with prolonged AI usage while maintaining interpretable reasoning traces.

Knowledge Graph Generation from Large Language Models

- Built an automated pipeline for generating structured knowledge graphs from unstructured text using large language models and evaluated GPT-4, LLaMA-2, and BERT for relation extraction and GraphRAG integration.

Time Series Foundation Model Evaluation for Anomaly Detection

- Benchmarked Time Series Foundation Models such as TimeGPT, Chronos, and Time-MOE across multiple datasets, showing that traditional statistical and deep learning models can outperform TSFMs for anomaly detection and prediction.

Med-Bot: Retrieval Augmented Medical Assistant

PUBLICATIONS

Leveraging LSTM for Predictive Modeling of Satellite Clock Bias

- *Presented at [CDMA 2025 conference](#) and available on [IEEE Xplore](#)*

Generating Knowledge Graphs from Large Language Models

Time Series Foundational Models: Their Role in Anomaly Detection and Prediction

- *Presented at [AAAI 2025 Anomaly Detection in Scientific Domains Workshop](#)*

Med-Bot: An AI-Powered Assistant to Provide Accurate and Reliable Medical Information